SYLLABUS : SEMICONDUCTOR

1. The forbidden energy band gap in conductors, semiconductors and insulators are EG_1, EG_2 and EG_3 respectively, The relation among them is

(A) $EG_1 = EG_2 = EG_3$ (B) $EG_1 < EG_2 < EG_3$ (C) $EG_1 > EG_2 > EG_3$ (D) $EG_1 < EG_2 > EG_3$

2. Fermi energy is the-

(A) Minimum energy of electrons in metal at 0 K

(B) Minimum energy of electrons in metal at 0°C

(C) Maximum energy of electrons in metal at 0 K

(D) Maximum energy of electrons in metal at 0°C

3. The valence band at (0 K) is-

(A) completely filled (B) completely empty (C) partially filled (D) nothing can be said

4. The free electron concentration (n) in the conduction band of a semiconductor at a temperature T Kelvin is described in terms of E_g and T as-

(A) $n = ATe^{-Eg/kT}$ (B) $n = AT^2e^{-Eg/kT}$ (C) $n = AT^2e^{-Eg/2kT}$ (D) $n = AT^{3/2}e^{-Eg/2Kt}$

5. Electric conduction in a semiconductor takes place due to

(A) electrons only (B) holes only

(C) both electrons and holes (D) neither electron nor holes

6. The mobility of free electron is greater than that of free holes because

(A) They carry negative charge

(B) They are light

(C) They mutually collide less

(D) They require low energy to continue their motion

7. Let n_p and n_e be the numbers of holes and conduction electrons in an intrinsic semiconductor

(A) $n_p > n_e$ (B) $n_p = n_e$ (C) $n_p < n_e$ (D) $n_p \neq n_e$

8. A N-type semiconductor is

(A) Negatively charged (B) Positively charged (C) Neutral (D) None of these

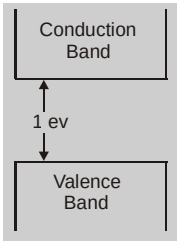
9. Let n_p and n_e be the numbers of holes and conduction electrons in an extrinsic semiconductor

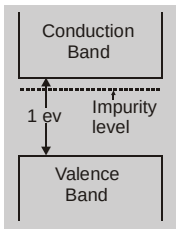
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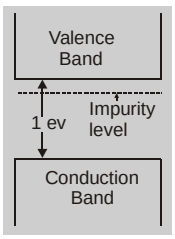
10. In a P-type semiconductor, the acceptor level is 57 meV, above the valence band. The maximum wave length of light required to produce a hole will be-

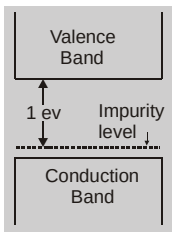
(A) 57 A^o (B) 57×10^{-3} A^o (C) 217100 A^o (D) 11.61×10 A^o

11. Which of the following energy band diagram shows the N-type semiconductor-

(A) 

(B) 

(C) 

(D) 

12. A P-type silicon semiconductor is made by adding one atom of indium per 5×10^7 atoms of silicon is 25×10^{28} atom/m.^{3}. Point the number of acceptor atoms in per cubic cm. of silicon

(A) 2×10^{30} atom/cm^3 (B) 5×10^{15} atom/cm^3

(C) 1×10^{15} atom/cm^3 (D) 2.5×10^{36} atom/cm^3

13. If N-type semiconductor is heated then-

(A) The number of electrons increases and the number of holes decreases

(B) The number of holes increases and the number of electrons decreases

(C) The number of electrons and holes both remain equal

(D) The number of both electrons and holes increases equally

14. GaAs is-

(A) an elemental semiconductor (B) a compound semiconductor

(C) an insulator (D) a metallic semiconductor

15. Forbidden energy gap of Ge is 0.75 eV, maximum wave length of incident radiation for producing electron-hole pair in germanium semiconductor is-

(A) 4200 A^o (B) 16500 A^o (C) 4700 A^o (D) 4000 A^o

16. Mobility of electrron in N-type Ge is 5000 cm^2 / volt sec and conductivity 5 mho/cm. If effect of holes is negligible then impurity concentration will be-

(A) 6.25×10^{15}/cm^3 (B) 9.25×10^{14}/cm^3 (C) 6×10^{13}/cm^3 (D) 9×10^{13} / cm^3

17. The intrinsic carrier density in germanium crystal at 300 K is 2.5×10^{13} per cm^3. If the electron density in an N-type germanium crystal at 300 K be 0.5×10^{17} per cm^3, the hole density (per cm^3) in this N-type crystal at 300 K would be expected around-

(A) 2.5×10^{13} (B) 5×10^6 (C) 1.25×10^{10} (D) 0.2×10^4

18. Pure Si at 300 K has equal electron (n_e) and hole (n_h) concentrations of 1.5×10^{16} m^{-3}. Doping by indium increases n_h to 4.5×10^{22} m^{-3}. Calculate n_e in the doped Si-

(A) 5.0×10^9 m^{-3} (B) 6.0×10^6 m^{-3} (C) 7.0×10^3 m^{-3} (D) 4.0×10^9 m^{-3}

19. The length of a germanium rod is 0.58 cm and its area of cross-section is 1mm^2. If for germanium $n_i = 2.5 \times 10^{19}$ m^{-3}, $\mu_h = 0.19$ m^2/ V-s, $\mu_e = 0.39$ m^2/ V-s, then the resistance of the rod will be -

(A) 2.5 KΩ (B) 5.0 KΩ (C) 7.5 KΩ (D) 10.0 KΩ

20. The contributions in the total current flowing through a semiconductor due to electrons and holes are $\frac{3}{4}$ and $\frac{1}{4}$ respectively. If the drift velocity of electrons is $\frac{5}{2}$ times that of holes at this temperature, then the ratio of the concentrations of electrons and holes is-

(A) 6 : 5 (B) 5 : 6 (C) 3 : 2 (D) 2 : 3

21. The P-N junction is-

(A) an ohmic resistance (B) an non ohmic resistance

(C) a positive resistance (D) a negative resistance

22. Diffusion current in a p-n junction is greater than the drift current in magnitude

(A) if the junction is forward-biased (B) if the junction is reverse-biased

(C) if the junction is unbiased (D) in no case

23. A hole diffuses from the p-side to the n-side in a p-n junction. This means that

(A) a bond is broken on the n-side and the electron freed from the bond jumps to the conduction band

(B) a conduction electron on the p-side jumps to a broken bond to complete it

(C) a bond is broken on the n-side and the electron freed from the bond jumps to a broken bond on the p-side to complete it

(D) a bond is broken on the p-side and the electron freed from the bond jumps to a broken bond on the n-side to complete it.

24. The depletion region of a P-N diode, under open circuit condition contains-

(A) Electrons (B) Holes

(C) Unmasked immobile impurity ions (D) Impurity atoms

25. Which is the wrong statement in following sentence? A device in which P and N type semiconductors

are used is more useful then a vacuum tube because-

(A) power is not necesary to heat the filament

(B) it is more stable

(C) very less heat is producted in it

(D) its efficiency is high due to a high voltage drop across the junction

ANSWER KEY

1.	(B)	**2.**	(C)	**3.**	(A)	**4.**	(D)	**5.**	(C)
6.	(D)	**7.**	(B)	**8.**	(C)	**9.**	(D)	**10.**	(C)
11.	(B)	**12.**	(B)	**13.**	(D)	**14.**	(B)	**15.**	(B)
16.	(A)	**17.**	(C)	**18.**	(A)	**19.**	(A)	**20.**	(A)
21.	(B)	**22.**	(A)	**23.**	(C)	**24.**	(C)	**25.**	(D)